MSE 311 Thermodynamics of materials-Homework 2

Name: *Student ID:*

1. Discuss *how the energy is conservation* when a man is walking. (Explain through the following formula)

$$dU = \delta q - \sum \delta w_i$$

2. One mole of an ideal gas at 25°C and 1 atm undergoes the following reversibly conducted cycle:

a. An isothermal expansion to 0.5 atm, followed by
b. An isobaric expansion to 100°C, followed by
c. An isothermal compression to 1 atm, followed by
d. An isobaric compression to 25°C.

The system then undergoes the following reversible cyclic process.

a. An isobaric expansion to 100°C, followed by
b. A decrease in pressure at constant volume to the pressure ***P*** atm, followed by
c. An isobaric compression at ***P*** atm to 24.5 liters, followed by
d. An increase in pressure at constant volume to 1 atm.

Calculate the value of ***P*** which makes the work done on the gas during the first cycle equal to the work done by the gas during the second cycle.

3. Ten moles of ideal gas, in the initial state P_1 = 10 atm, T_1 = 300 K, are taken round the following cycle:

a. A reversible change of state along a straight line path on the *P*-*V* diagram to the state P = 1 atm, T = 300 K,
b. A reversible isobaric compression to V = 24.6 liters, and
c. A reversible constant volume process to P = 10 atm.

How much work is done on or by the system during the cycle? Is this work done on the system or by the system?

4.Open question. PV^{-1}=const in the balloon?